

Track 4 · Shopping Copilot · Astrid CLI Engine · Zero Token Cost

ShopPilot

Offline-first multi-turn shopping copilot

TechnicalScore: 0.907

Hit@10: 97.5%

MRR: 0.858

MTTC: 2.88 turns

github.com/algorathem/techjam2026-shopping-copilot

Why Static Keyword Retrieval Breaks in Conversational Search

Single-turn search assumes fixed intent. Real shopping sessions suffer from four core failure modes:

01

Vague queries

"something nice for summer" floods the returned results

02

Semantic Collisions

"dress sandals" pulling dresses instead of footwear

03

Mind-change

Mid-session pivots leave stale context in retrieval

04

Turn Fatigue

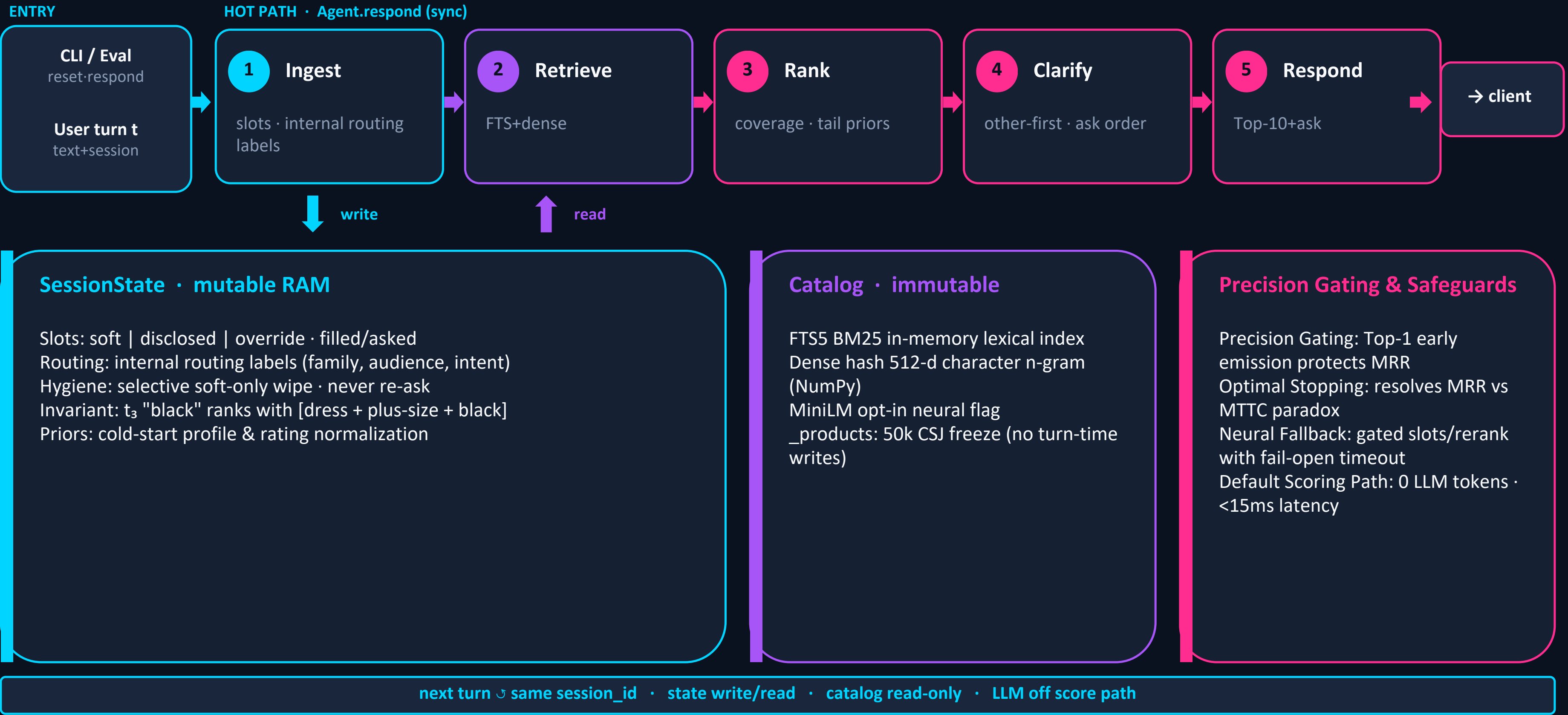
miss after turn 10 → cart abandonment

Objective Function

Find the hidden parent_asin within a 50k frozen catalog under the official formula:

$$\text{TechnicalScore} = 0.50 \cdot \text{Hit@10} + 0.30 \cdot \text{MRR} + 0.20 \cdot \left(\frac{11 - \text{MTTC}}{10} \right)$$

ShopPilot Engine: Headless, Fast, and Offline



Public 200 — weak BM25 vs ShopPilot

Evaluated deterministically via official evaluator.local_evaluator on the 50,000 Amazon CSJ dataset.

TechnicalScore

0.907

from 0.107 · ~8.5×

Hit@10

0.975

from 0.125

MRR

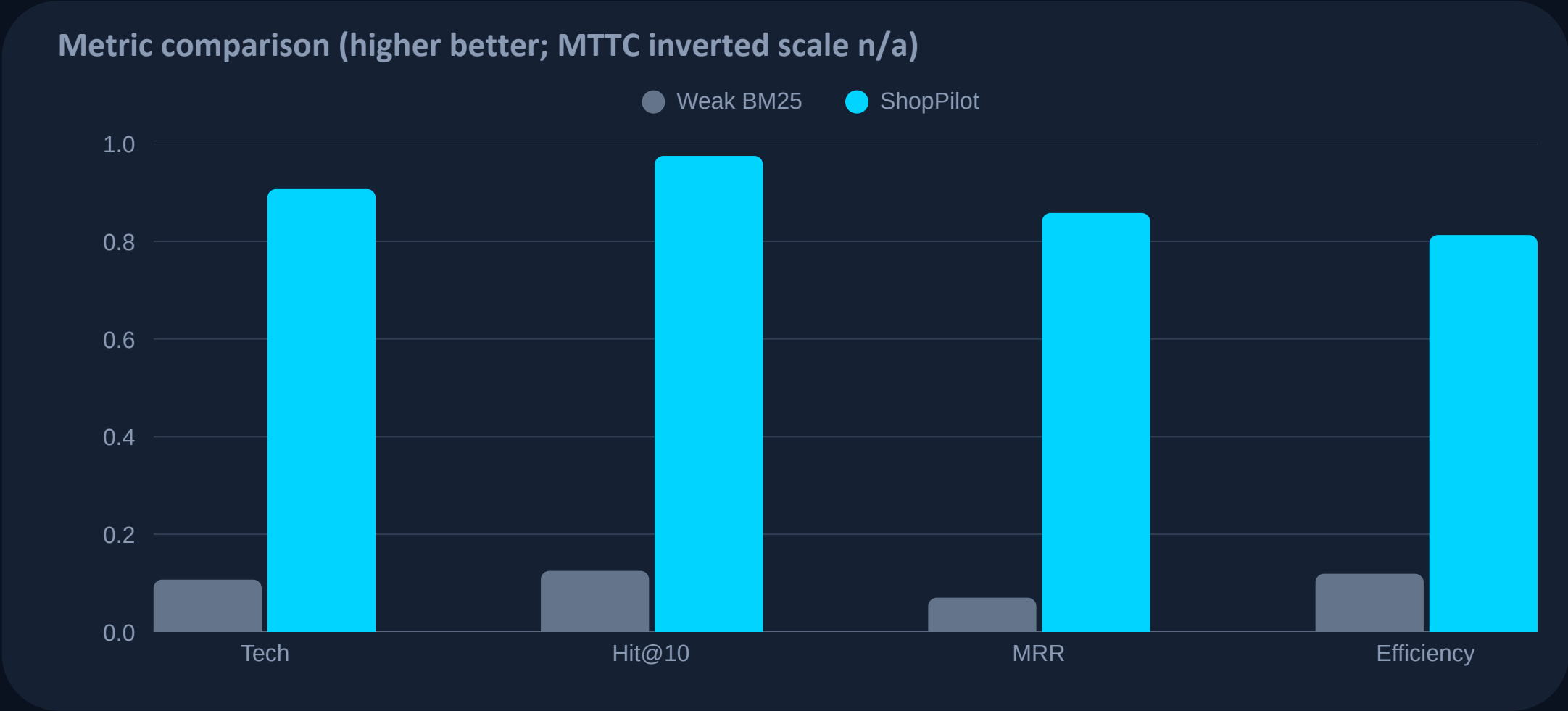
0.858

from 0.068

MTTC

2.88

from 9.81 turns



METRIC TRADE-OFF · THE EARLY-TERMINATION PARADOX

Emitting Top-10 too early on Turn 1 risks a low-rank hit (e.g. Rank 8 MRR=0.125), freezing a poor score. Because MRR carries 1.5× the weight of MTTC in TechnicalScore, ShopPilot uses Precision Gating (emits only Top-1 on ambiguous early turns) to gather constraints and force final conversion toward Rank 1.

Meet Astrid: ShopPilot in Action



Astrid · quiet clarity for every aisle
offline hybrid shopping agent · dense=hash
commands: /new /state /quit

START COMMAND

python3 cli-chat.py
github.com/algorathem/techjam2026-shopping-copilot